

Tablet-Based Quality Inspection & Reporting System

Functional & Technical Requirements Document

Project Name: Manufacturing Quality Control System

Document

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Target Audience: Engineering, Software Development, QA Teams

1. Project Objective & Scope

This document outlines the system and functional requirements for a tablet-optimized shop floor Quality Inspection System. The purpose of this system is to replace manual paper-based logs with a robust, real-time digital entry screen. This ensures continuous adherence to product engineering specifications, delivers instantaneous pass/fail visual cues to floor inspectors, automates immediate reporting, and facilitates simple master data management via bulk Excel uploads.

2. Functional Requirements

2.1. Tablet-Optimized Inspection Entry

The system must feature a specialized touch-friendly interface optimized for 10-inch tablets used by inspectors directly at work centers.

- **Part and Operation Selection:** The user must select a Part Number from a searchable dropdown list. Upon selection, the system must filter and dynamically display only the associated Operation Numbers.
- **Parameter Fetching:** Selecting both values will load all engineering dimensions, parameters, and their specific nominal and upper/lower tolerance values to be audited.
- **Touch-friendly Controls:** Dropdowns, text inputs, and submit buttons must have a height profile appropriate for rapid fingertip entry without precise cursor devices.

2.2. Real-Time Tolerance Validation & Visual Cues

To eliminate manual errors and processing delays, data validation occurs locally and instantly upon value input.

- **Instantaneous Validation:** Immediately upon entering a value or moving out of focus from a parameter numeric input box, the system compares the value to the defined upper and lower tolerance limits.
- **Visual Fail Status Indicator:** If the measurement falls outside the allowable boundaries, the specific cell or field must instantly transition to a high-visibility alert design (e.g., bright red fill, bold **FAIL** text badge).
- **Visual Pass Status Indicator:** Valid measurements must show a subtle green background or checkmark signifying acceptance.

CRITICAL RULE: Visual Feedback Lock

If any single checked parameter triggers a "FAIL" state, the final submitted inspection sequence must flag the entire lot number as "Rejected" within the tracking database automatically.

2.3. Conceptual Layout (Tablet UI)

Below is a visual layout concept for the entry screen interface designed for tablet form factors:

Quality Entry Portal — Shop Floor Tablet View

Part Number:

PART-9904-X
(Selected)

Operation No:

OP-20: CNC Milling

Parameter Name	Specification / Tolerance	Observed Value	Status
Outer Diameter (OD)	45.00mm ± 0.05mm [44.95 - 45.05]	45.02	PASS
Wall Thickness	12.50mm + 0.10mm / -0.00mm	12.62	FAIL

3. Automated Inspection Report Generation

Upon clicking the **"Save / Submit"** option on the tablet screen, the system writes entries directly to the production log database and instantly processes a professional compliance report.

- **Instant Output:** A preview of the inspection report is rendered on screen immediately for the inspector.
- **Format Requirements:** Reports are downloadable as standardized PDFs or print-ready layouts incorporating company metadata, timestamp, user signature block, complete parametric run histories, and highlighted out-of-spec items.

4. Master Data Management & Bulk Upload

To avoid manually adding individual configurations via a web form, administrators can manage engineering standards using bulk imports.

- **Excel Template Provisioning:** The administration console provides downloadable standardized .xlsx templates.
- **Data Validation during Import:** The data engine will validate the layout, verify that fields are populated correctly (e.g., text vs numeric tolerances), and alert users of any overlapping rules before committing records.

Recommended Master Data Excel Schema

Column A	Column B	Column C	Column D	Column E	Column F
Part_Number	Operation_Number	Parameter_Name	Nominal_Value	Lower_Tolerance	Upper_Tolerance
PART-9904-X	OP-20	Outer Diameter (OD)	45.00	-0.05	0.05
PART-9904-X	OP-20	Wall Thickness	12.50	0.00	0.10

5. Technical & Non-Functional Specifications

Attribute	Requirement Specification
Tablet Compatibility	Responsive layout optimized specifically for Apple iPad (iOS) and Android tablets with screen sizes ≥ 10.1 inches. Native cross-browser compliance (Safari, Chrome).
Response Latency	Real-time tolerance evaluation must execute within < 200 milliseconds from input loss-of-focus. Reports must render within < 2 seconds from form submission.
Offline Support	Local storage queuing capacity to log entries in case of momentary Wi-Fi dropouts on the floor, syncing automatically once connectivity resets.